



Software Requirements Specification (SRS)

Software Development, AI, Database and Cyber Security Division

Project title: OmniScan the digital intelligence engine

Prepared by: Development team

Table of Contents

Table of Contents	i
List of Tables	ii
List of Figures	iii
Executive Summary	iv
1. Introduction	1
1.1 Project Overview	1
1.2 Business Context	1
1.3 Purpose of this Document	1
1.4 Scope	1
2. Project Information	2
3. Objectives	3
3.1 Primary Objective	3
3.2 Specific Goals	3
3.3 Success Metrics	3
4. Feasibility Study	3
4.1 Technical Feasibility	4
4.2 Operational Feasibility	4
4.3 Schedule Feasibility	4
4.4 Economic Feasibility	4
5. Stakeholders & Actors	5
6. Functional Requirements	6
7. Use Case Descriptions	7
8. Core Data Fields (PO Record)	10
9. Non-Functional Requirements	11
10. Constraints & Assumptions	12
10.1 Constraints	12
10.2 Assumptions	12
11. Development Schedule	13
12. Testing Plan	14
13. Glossary	15
14. References	16

List of tables

Table 1 Project information.....	2
Table 2 Feasibility Study.....	4
Table 3 Stakeholders and Actors.....	5
Table 4 Functional Requirements.....	6
Table 5 Use Case Descriptions.....	7
Table 6 Core Data Fields(PO records).....	8
Table 7 Non-Functional Requirements.....	9
Table 8 Development Schedule.....	11
Table 9 Testing plan.....	12
Table 10 Glossary.....	13

List of Figures

Diagram 1 Use Case Diagram.....	8
Diagram 2 Squence Diagram.....	9

Executive Summary

OmniScan :- The Digital Intelligence Engine is a web-based document digitization and management system developed for **LATA AGRI EXPORT division of ALTA Computec PLC**. The system replaces manual, paper-based handling of international Purchase Order (PO) documents with an efficient, AI-powered digital workflow.

LATA AGRI EXPORT regularly processes PO confirmations from international buyers such as EFICO NV. These documents contain critical trade data, including quantities, pricing (FOB), shipment schedules, compliance requirements (EUDR), and payment terms. Currently, this information is managed manually, creating risks such as data loss, delayed shipments, and limited traceability.

OmniScan addresses these challenges by enabling staff to capture documents using smartphones or upload digital files, automatically extract structured data using AI-powered OCR, and store all records in a centralized, searchable archive. The system also provides automated deadline notifications, compliance tracking, and role-based access control to ensure operational efficiency and accountability.

Key functionalities include intelligent data extraction, full-text search, shipment and payment deadline alerts, EUDR compliance monitoring, report generation (PDF/Excel), and pre-shipment sample tracking. These features significantly reduce manual effort while improving accuracy and decision-making.

The system is built using modern technologies, including React for the frontend, Python for the backend, and PostgreSQL or MySQL for data storage, integrated with AI services for document processing. It requires no specialized hardware, making it cost-effective and easy to deploy within the existing infrastructure.

Development follows a 10-week Agile model, ensuring incremental delivery and early usability of core features. The expected outcomes include high OCR accuracy (approximately 90% on clear images), fast document retrieval, reliable deadline tracking, and rapid user adoption.

This SRS serves as a formal agreement between stakeholders and the development team, clearly defining system scope, both functional & non-functional requirements, assumptions, and constraints. It provides a structured foundation for system design, development, testing, and deployment.

1. Introduction

1.1 Project Overview

OmniScan is a web-based document digitization and management system specifically designed for LATA AGRI EXPORT, a department within ALTA Computec PLC and an Ethiopian green coffee export company. The system enables staff to capture, store, search, and manage physical purchase order (PO) documents — such as those issued by international buyers like EFICO NV (Antwerpen, Belgium) — by photographing or scanning them using any smartphone, tablet, or desktop device.

The system eliminates manual paper-based filing of export contracts, automates key data extraction (buyer, seller, quantity, price, shipment dates), and provides real-time alerts for upcoming shipment deadlines and payment due dates.

1.2 Business Context

LATA AGRI EXPORT regularly receives international Purchase Order confirmations (e.g., PO EFI25120039 from EFICO NV for 160 bags of Ethiopia Djimmah Grade 5 green coffee). These documents contain critical trade data including:

- Buyer and seller information
- Coffee quantity, quality description, and crop year
- FOB price per unit (e.g., USC 295.00/LB)
- Shipment period and origin port (e.g., Djibouti)
- Destination port (e.g., Antwerp, Belgium)
- Payment terms (net cash against documents)
- EUDR compliance requirements
- Special pre-shipment sample conditions

Currently these documents are managed manually, creating risks of lost records, missed shipment deadlines, and difficulty searching historical contracts.

1.3 Purpose of this Document

This SRS defines the complete functional and non-functional requirements for OmniScan, including system scope, stakeholder roles, use cases, constraints, and development timeline. It serves as the contract between LATA AGRI EXPORT and the development team.

1.4 Scope

OmniScan will cover: document capture via camera/file upload, AI-assisted data extraction, searchable document archive, shipment and payment deadline notifications, reporting and export, and role-based access for LATA staff.

2. Project Information

Field	Details
Project Title	OmniScan — The Digital Intelligence Engine
Client Organization	LATA AGRI EXPORT (ALTA Computec PLC)
Client Address	Addis Ababa, Ethiopia
Industry	Green Coffee Export (International Trade)
Primary Use Case	Digitize EFICO-style purchase order documents from photo or scan
Tech Stack	React (frontend), Python(backend), MySQL/PostgreSQL (DB), AI API (OCR/extraction)
Development Model	Agile — 5 sprints over 10 weeks
Estimated Effort	30 hours/week (team of 3)

3. Objectives

3.1 Primary Objective

To provide LATA AGRI EXPORT with a reliable, easy-to-use digital archive for all purchase order documents, replacing paper-based filing and enabling instant search, automated data extraction, and proactive deadline notifications.

3.2 Specific Goals

- Enable staff to capture PO documents by photographing them with a phone or uploading PDF/image files
- Automatically extract structured data (buyer, seller, quantity, price, dates) using AI/OCR
- Store all documents in a searchable, filterable digital archive
- Send email/SMS notifications for approaching shipment and payment deadlines
- Generate export reports (PDF/Excel) of active POs by date range, buyer, or crop
- Support role-based access: Admin, Trade Manager, Data Entry Operator

3.3 Success Metrics

- 95% OCR extraction accuracy on clear document photos
- Document search returns results in under 2 seconds
- 100% of shipment deadlines flagged at 14-day and 3-day intervals
- 80% staff adoption within first month of deployment

4. Feasibility Study

4.1 Technical Feasibility

The project uses proven, widely-adopted technologies: React for a responsive browser-based interface accessible on phones and desktops, Python/Flask for the backend API, and AI-assisted OCR for document text extraction. No specialized hardware is required — any smartphone camera can capture PO documents. Cloud storage (e.g., Firebase Storage or AWS S3 free tier) handles document images.

4.2 Operational Feasibility

Staff at LATA AGRI EXPORT are already familiar with smartphones and web browsers. The capture workflow (open app → photograph document → review extracted data → save) requires minimal training. An onboarding tutorial and help tooltips will be embedded in the application.

4.3 Schedule Feasibility

The 20-week Agile plan (5 sprints × 4 weeks) includes buffer weeks and iterative releases, allowing LATA to start using core features from Sprint 3 onward while advanced features (notifications, reporting) are completed in Sprints 4–5.

4.4 Economic Feasibility

Cost Item	Estimated Cost (ETB)	Notes
Development (in-house)	0	Internal team
Internet & hosting (12 months)	6,000 – 12,000	Cloud free tiers or VPS
SMS notification service	500 – 2,000	Per notification volume
Training & onboarding	500 – 1,000	Tutorial materials
Total Estimated	7,000 – 15,000 ETB/year	Highly cost-effective

Overall Feasibility: HIGH — the system delivers significant time savings and risk reduction relative to minimal cost.

5. Stakeholders & Actors

Actor	Role	Permissions
System Admin	Full system control	Manage users, database backup, settings
Trade Manager	Oversees all POs and reports	View/search all POs, approve, generate reports
Data Entry Operator	Captures and indexes documents	Upload, capture, extract, edit PO records
Finance Officer	Tracks payment status	View finance data, payment status, reminders
External System (EFICO)	Issues PO documents	No system access — documents ingested by upload

6. Functional Requirements

ID	Requirement	Priority	
FR-01	User Authentication: Login with email/password; role-based access control (RBAC)	Mandatory	
FR-02	Document Capture: Upload photo (JPG/PNG) or PDF of PO document via mobile or desktop browser	Mandatory	
FR-03	AI Data Extraction: Automatically extract buyer, seller, quantity, quality, price, FOB terms, shipment dates, destination from uploaded image using OCR + AI	Mandatory	
FR-04	Manual Correction: Allow staff to review and edit extracted data before saving	Mandatory	
FR-05	Document Archive: Store original image + extracted structured data; support pagination and sorting	Mandatory	
FR-06	Search & Filter: Full-text search by buyer name, PO number, date range, shipment port, crop year	Mandatory	
FR-07	Shipment Deadline Alerts: Notify responsible staff via email/browser notification 14 days and 3 days before shipment start and end dates	Mandatory	
FR-08	Payment Status Tracking: Record payment status (Pending/Received/Overdue) per PO; trigger overdue reminders	Mandatory	
FR-09	Pre-Shipment Sample Tracking: Flag POs requiring pre-shipment samples with due date (1 month before shipment)	Mandatory	
FR-10	EUDR Compliance Flag: Mark each PO's EUDR status	Mandatory	
FR-11	Report Generation: Export filtered PO list as PDF or Excel by date range, buyer, or status	Mandatory	
FR-12	Audit Log: Track who uploaded, edited, or deleted each document record	Mandatory	
FR-13	Dashboard: Show summary cards (total active POs, upcoming shipments, overdue payments, POs pending samples)	Mandatory	
FR-14	Bulk Import: Upload multiple PO documents at once (batch processing)	Optional	
FR-15	Document Versioning: Maintain version history if a PO document is replaced	Optional	

7. Use Case Descriptions

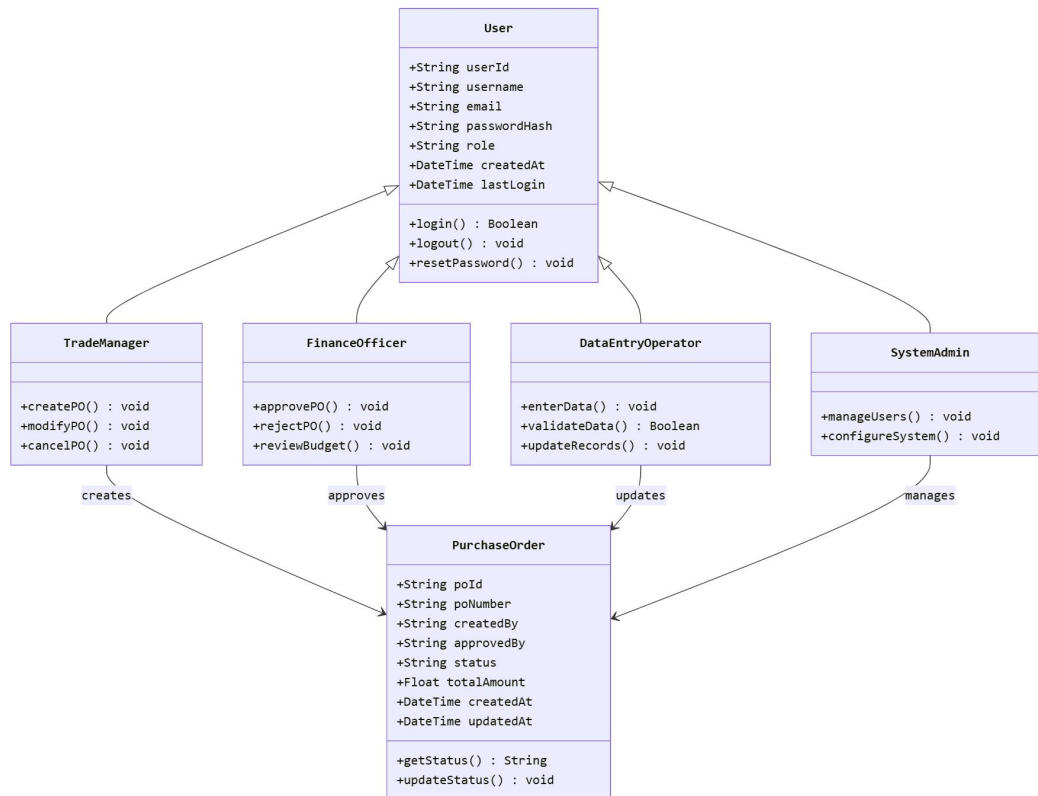
ID	Use Case	Actor	Basic Flow	Postcondition
UC-01	Capture PO Document	Data Entry Operator	Open capture screen → take photo or upload file → system processes image → extracted data shown for review → confirm & save	PO record created in archive
UC-02	Search PO Archive	All Users	Enter search term or filter criteria → system queries DB → results displayed with thumbnails → click to view full record	Relevant POs displayed
UC-03	Review & Edit Extracted Data	Data Entry Operator	After capture, view pre-filled form → correct any OCR errors → attach tags → save record	Accurate PO data stored
UC-04	View PO Dashboard	Trade Manager	Login → dashboard shows summary stats → click any card to drill down into list	Overview of all active POs
UC-05	Receive Deadline Notification	Trade Manager, Finance Officer	System checks daily → finds POs with shipment date within 14 or 3 days → sends email notification with PO details	Staff notified of upcoming deadline
UC-06	Update Payment Status	Finance Officer	Open PO record → update Payment Status field → system records change with timestamp	Payment status updated
UC-07	Generate Report	Trade Manager	Select date range & filters → click Export → system generates PDF/Excel file → file downloaded	Report file downloaded
UC-08	Manage Users	System Admin	Open user management → add/edit/deactivate user accounts → assign roles	User accounts updated
UC-09	Track EUDR Compliance	Data Entry Operator	On PO record → set EUDR Approach and Engagement Level fields → system flags non-compliant records	EUDR status recorded
UC-10	Flag Pre-Shipment Sample	Trade Manager	System auto-calculates sample due date (1 month before shipment) → creates task → staff marks as Sent	Sample dispatch tracked

This use case diagram shows how users interact with the system to scan, upload, and manage documents.

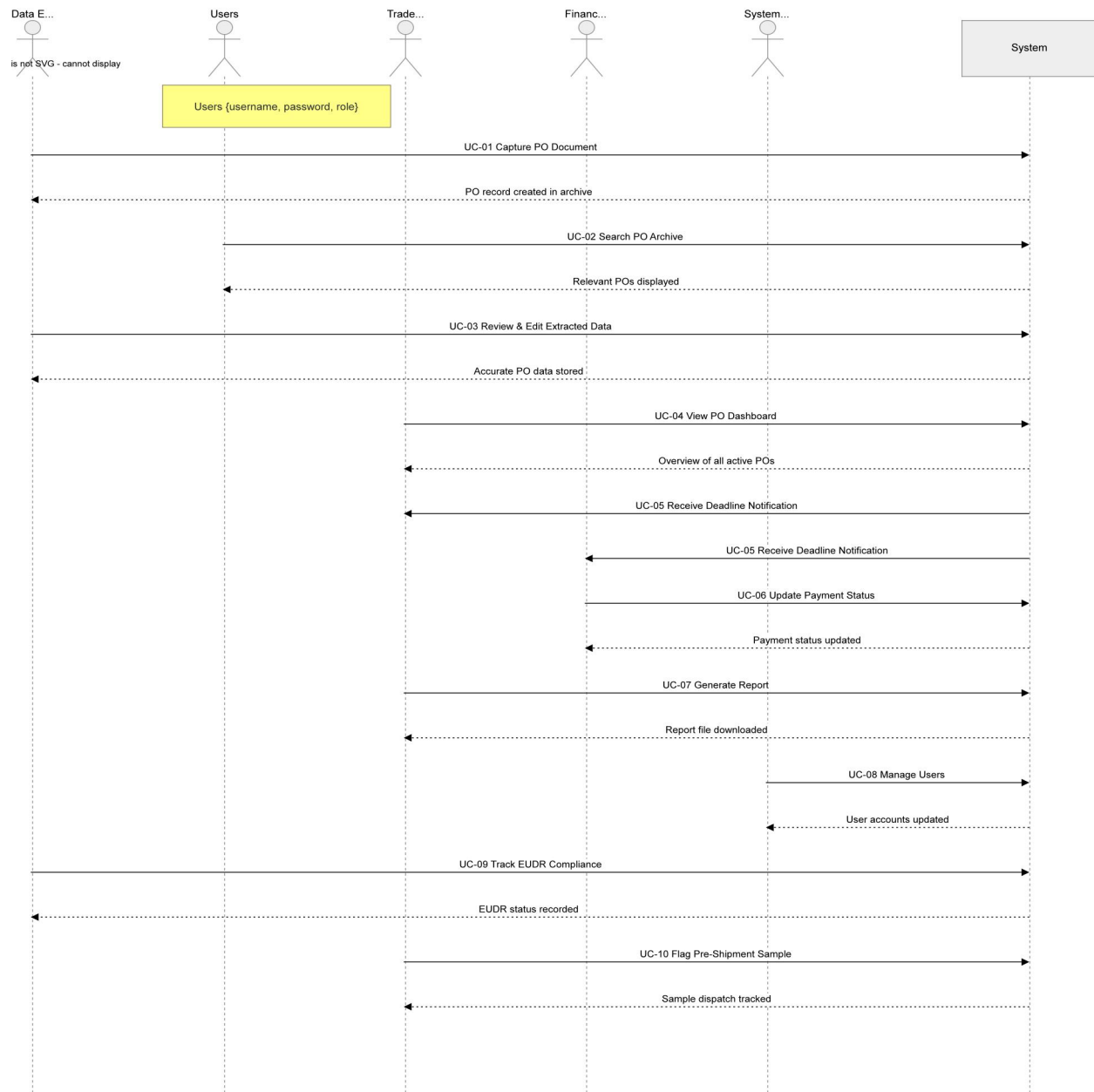
The system automatically processes documents using AI for text extraction, categorization, and storage.

Admins manage users, data control, and system configurations.

Overall, it illustrates a smart workflow for converting scanned documents into searchable digital records.



This diagram illustrates the workflow of the system, showing how data moves from input through processing to final storage and retrieval. It highlights the interaction between key components such as capture, categorization, and database management. Overall, it provides a clear high-level view of how the system ensures efficient document handling and access.



8. Core Data Fields (PO Record)

Each digitized Purchase Order will store the following structured fields, based on the EFICO confirmation format:

Field Name	Data Type	Source	Example
po_number	String	OCR / Manual	EFI25120039
po_date	Date	OCR / Manual	05/12/2025
buyer_name	String	OCR / Manual	Efico n.v.
buyer_address	String	OCR / Manual	Italielei 181, B-2000 Antwerpen
seller_name	String	OCR / Manual	LATA AGRI EXPORT
quantity_bags	Integer	OCR / Manual	160
bag_weight_kg	Float	OCR / Manual	60
quality_description	String	OCR / Manual	Ethiopia Djimmah GR 5 Crop 2024/2025
price_per_unit	Float	OCR / Manual	295.00
price_currency	String	OCR / Manual	USC (US Cents)
price_unit	String	OCR / Manual	LB
incoterms	String	OCR / Manual	FOB Djibouti, Incoterms 2020
shipment_start	Date	OCR / Manual	03/12/2025
shipment_end	Date	OCR / Manual	31/01/2026
origin_port	String	OCR / Manual	Djibouti
destination_port	String	OCR / Manual	Antwerp, Belgium
payment_terms	String	OCR / Manual	Net cash against documents
payment_status	Enum	Manual	Pending / Received / Overdue
eudr_approach	Enum	Manual	Full Chain of Custody
eudr_engagement	String	Manual	Level 1: Geolocation Data Only
sample_required	Boolean	Rule	True/False
sample_due_date	Date	Calculated	Auto: 1 month before shipment start
sample_sent	Boolean	Manual	True/False
document_image_url	URL	System	Stored in cloud storage
uploaded_by	String	System	User ID
created_at	DateTime	System	Auto-generated

9. Non-Functional Requirements

Category	Requirement
Security	RBAC with JWT tokens; all document images stored with access control; passwords hashed (bcrypt); HTTPS enforced; audit log of all data changes
Performance	Search results returned in <2 seconds; document upload and OCR processing <30 seconds per document; dashboard loads in <3 seconds
Usability	Mobile-first design; PO capture workflow completable in <5 taps on phone; onboarding tutorial for new staff; English + Amharic UI labels
Reliability	99% uptime; daily automated database backup; graceful offline handling (cache last viewed records)
Scalability	Support up to 5,000 PO records and 50 concurrent users without performance degradation
OCR Accuracy	AI extraction accuracy $\geq 95\%$ for clear document photos taken in adequate lighting
Notification Delivery	>99% delivery rate for email deadline notifications
Maintainability	Modular codebase (MVC pattern); >75% unit test coverage; API documentation; Git version control with CI
Compliance	EUDR compliance tracking per EU Regulation 2023/1115; data privacy per Ethiopian data protection norms
Browser Compatibility	Chrome, Firefox, Safari, Edge (desktop and mobile)

10. Constraints & Assumptions

10.1 Constraints

- System is browser-based only — no native Android/iOS app in v1
- OCR accuracy depends on photo quality — blurry or dark images may require manual correction
- Maximum file upload size: 20MB per document image
- Email notifications require active internet connection
- Initial deployment supports English UI only; Amharic labels added in v1.1

10.2 Assumptions

- Staff have smartphones with cameras capable of capturing readable document photos
- LATA AGRI EXPORT can provide a company email address for notification delivery
- Internet access is available at the office for uploads and searches
- EFICO and other buyers will continue using similar PO document formats

11. Development Schedule

Sprint	Weeks	Dates	Key Deliverables
Pre-Dev: Research & SRS	1	May 01–07, 2026	Requirements gathering, SRS finalization, UI wireframes
Sprint 1: Setup & Auth	2–3	May 07 –May14, 2026	Project setup, login/RBAC system, user management, database schema
Sprint 2: Document Capture & OCR	4–5	May14– May21, 2026	Camera/upload UI, AI OCR integration, data extraction form, document storage
Sprint 3: Archive & Search	6–7	May 21 – May 28, 2026	Searchable archive, filter/sort, PO detail view, dashboard summary cards
Sprint 4: Notifications & Finance	8–9	May28 –Jul05, 2026	Deadline email alerts, payment status tracking, sample tracking, EUDR flags
Sprint 5: Reports, Testing & Launch	9-10	Jul05 – Jul12, 2026	PDF/Excel export, UAT with LATA staff, bug fixes, deployment, training

12. Testing Plan

Testing Type	What It Tests	When
Unit Testing	Individual functions: OCR extraction accuracy, date calculation, search query logic	Every sprint
Integration Testing	API endpoints, OCR service connection, email notification delivery	Sprint 3–4
System Testing	End-to-end capture → extract → store → search → notify workflow	End of Sprint 4
UAT (User Acceptance)	Real LATA staff test with actual EFICO PO documents; 80% satisfaction target	Sprint 5
Performance Testing	Upload and OCR under concurrent users; search with 1,000+ records	Sprint 5
Security Testing	RBAC enforcement, unauthorized access attempts, file upload validation	Sprint 5

13. Glossary

Term	Definition
PO	Purchase Order — official document confirming a trade transaction
FOB	Free On Board — seller responsible for goods until loaded onto ship at origin port
EUDR	EU Deforestation Regulation (EU 2023/1115) — requires supply chain traceability for products entering EU
ESCC	European Standard Contract for Coffee — standard contract framework used by European Coffee Federation
OCR	Optical Character Recognition — technology to convert document images into machine-readable text
RBAC	Role-Based Access Control — permissions assigned based on user role
UAT	User Acceptance Testing — end-user validation of system functionality
SRS	Software Requirements Specification — this document
USC	United States Cents — pricing unit commonly used in coffee commodity trading
MVP	Minimum Viable Product — core features required for first usable release

14. References

- EU Regulation 2023/1115 on Deforestation-free Products (EUDR)
 - European Standard Contract for Coffee (ESCC) — European Coffee Federation
 - LATA AGRI EXPORT company profile and existing trade document formats
 - IEEE Std 830-1998 — IEEE Recommended Practice for Software Requirements Specifications
 - International Coffee Organization (ICO) trade standards
-